Essays

|!|BCEEB0080000001|!|
Describe and explain how oxygen in the atmospheric air passes into the lungs of humans and how it is absorbed into the blood. (11 marks)

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Oxygen is brought to the air sacs by inhalation (Max. 4)

- The intercostal muscles contract, and the rib cage moves upwards and outwards. (1)
- The diaphragm muscles contract, and the diaphragm becomes flattened. (1)
- The thoracic volume increases and the lungs expand. (1)
- The air pressure in the lungs becomes lower than the atmospheric pressure. (1)
- Atmospheric air, which contains a higher concentration of oxygen, is drawn into lungs. (1)

Gas exchange in the air sacs (Max. 4)

- In the air sacs of the lungs, oxygen in the incoming air dissolves in the water film on the inner surface of the air sacs. (1)
- As the oxygen concentration of the water film is higher than the blood of the capillaries, (1)
  oxygen diffuses across the walls of the air sacs and capillaries into the blood. (1)
- Oxygen then diffuses into red blood cells (1)
  and combines with haemoglobin in the red blood cells to form oxyhaemoglobin. (1)

Communication (Max. 3)

(11 marks)

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Skills assessed	Marks
Understanding basic concepts	8
Communication	3

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Describe how carbon dioxide is transported from a body cell to an air sac in the lungs and then removed by the lungs. (12 marks)

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Transport of carbon dioxide from the body cells to the lungs (Max. 3)

- Carbon dioxide is formed in body cells during respiration. (1)
- Carbon dioxide diffuses into the blood along the concentration gradient. (1)
- Most carbon dioxide enters the red blood cells and reacts readily with water to form hydrogencarbonate ions. (1)
- The ions diffuse out of the red blood cells and dissolve in the blood plasma. (1)
- Blood is then transported back to the heart via venae cavae and is then pumped to the lungs via the pulmonary artery. (1)

(Or other acceptable answers)

Gas exchange in the air sacs (Max. 2)

- The carbon dioxide concentration in the air sacs is lower than that in the blood. (1)
- When the blood reaches the lungs, hydrogencarbonate ions enter the red blood cells and break down to form carbon dioxide and water. (1)
- Carbon dioxide diffuses out of the blood plasma into the air of an air sac. (1)

Removal of carbon dioxide from the lungs by exhalation (Max. 4)

- The intercostal muscles relax, and the rib cage moves downwards and inwards. (1)
- The diaphragm muscles relax, and the diaphragm returns to dome-shaped. (1)
- The thoracic volume and lung volume decrease. (1)
- The air pressure in the lungs becomes higher than the atmospheric pressure. (1)
- Air in the air sacs, which contains a higher concentration of carbon dioxide, is forced out of the lungs. (1)

Communication (Max. 3)

(12 marks)

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Skills assessed	Marks
Understanding basic concepts	9
Communication	3

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